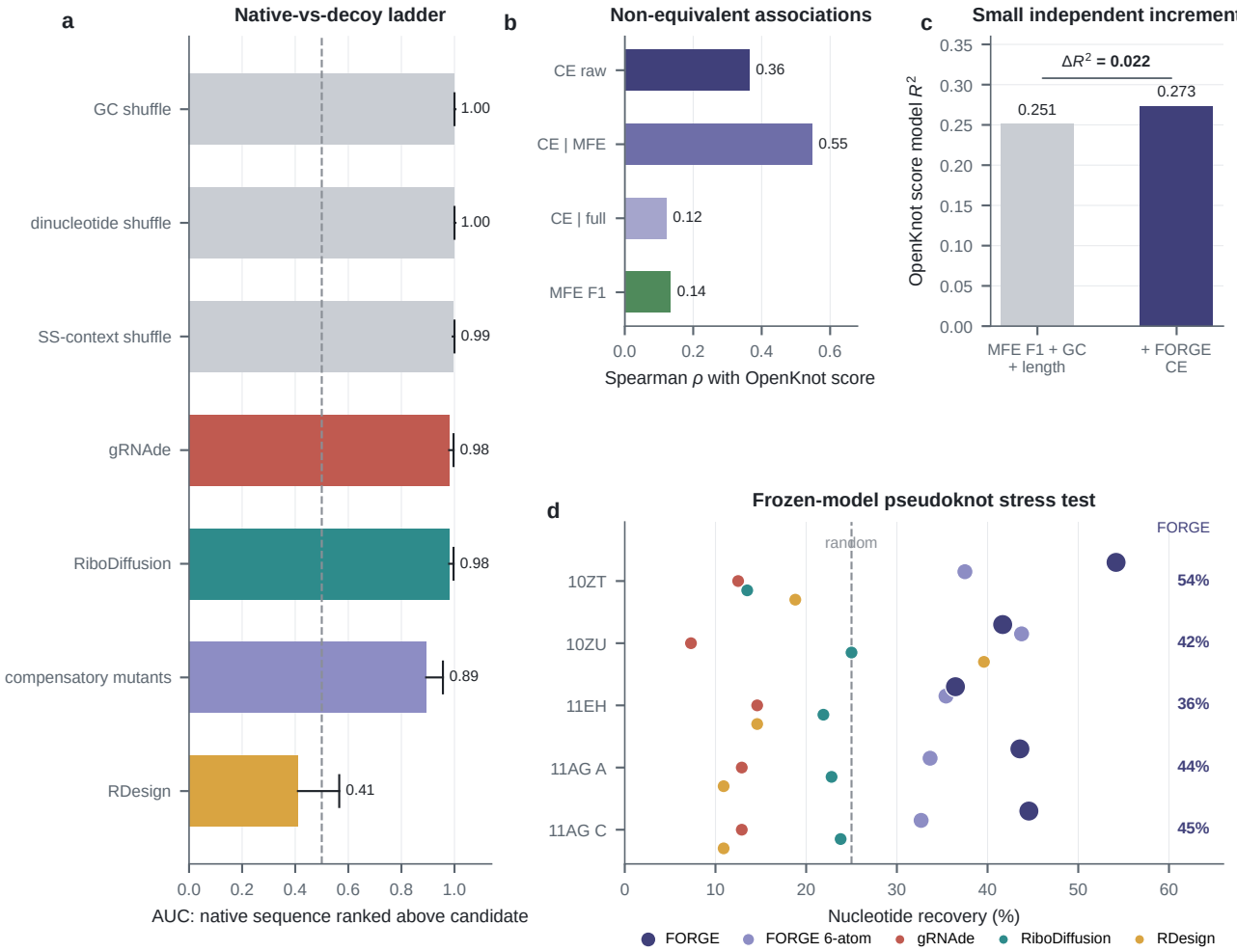
a
Native-vs-decoy ladder
GC shuffle
dinucleotide shuffle
SS-context shuffle
gRNAde
RiboDiffusion
compensatory mutants
RDesign
1.00
1.00
0.99
0.98
0.98
0.89
0.41
0.0
0.2
0.4
0.6
0.8
1.0
AUC: native sequence ranked above candidate
b
Non-equivalent associations
CE raw
CE | MFE
CE | full
MFE F1
0.36
0.55
0.12
0.14
0.0
0.2
0.4
0.6
Spearman ρ with OpenKnot score
c
Small independent increment
ΔR² = 0.022
0.251
0.273
OpenKnot score model R²
0.35
0.30
0.25
0.20
0.15
0.10
0.05
0.00
MFE F1 + GC + length
+ FORGE CE
d
Frozen-model pseudoknot stress test
random
FORGE
10ZT
10ZU
11EH
11AG A
11AG C
54%
42%
36%
44%
45%
0
10
20
30
40
50
60
Nucleotide recovery (%)
FORGE
FORGE 6-atom
gRNAde
RiboDiffusion
RDesign